OEIS A111114: An Exact Descent Criterion and Unconditional Growth Results

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Abstract

For $a(n) = \lfloor p_n/\pi(n) \rfloor$, $n \geq 2$, we prove an exact integer criterion for a decrease, the asymptotic $a(n) \sim (\log n)^2$, infinitely many increases, and density zero of the descent indices. These supporting results do not prove the OEIS conjecture that decreases occur infinitely often.

1 Definitions and exact criterion

Let p_n be the nth prime, let $\pi(x)$ count primes at most x, and define $a(n) = \lfloor p_n/\pi(n) \rfloor$ for integers $n \geq 2$, as in A111114. Put

$$k = \pi(n), \quad g = p_{n+1} - p_n, \quad t = \left\lfloor \frac{p_n}{k} \right\rfloor, \quad r = p_n - kt, \quad 0 \leq r < k.$$

Theorem 1. *If $n+1$ is composite, $a(n+1) \geq a(n)$. If $n+1$ is prime, then*

$$a(n+1) - a(n) = \left\lfloor \frac{r + g - t}{k+1} \right\rfloor.$$

In particular a decrease occurs exactly when $n+1$ is prime and $r + g < t$.

Proof. If $n+1$ is composite then $\pi(n+1) = k$ and the numerator increases. If $n+1$ is prime, its prime count is $k+1$ and

$$p_{n+1} = kt + r + g = (k+1)t + (r + g - t).$$

Division by $k+1$ and taking the floor proves the formula. The floor on the right is negative exactly when its numerator is negative. □

This includes $n = 2$: $k = 1, t = 3, r = 0, g = 2$, so the difference is -1.

2 Growth and frequency consequences

Theorem 2. *As $n \to \infty$, $a(n) \sim (\log n)^2$. There are infinitely many n with $a(n+1) > a(n)$. If $D(N)$ counts the integers $2 \leq n \leq N$ with $a(n+1) < a(n)$, then*

$$D(N) \leq \pi(N+1) - 1 = O\left(\frac{N}{\log N}\right).$$

In particular the descent indices have natural density zero.

Proof. The prime number theorem gives $p_n \sim n \log n$ and $\pi(n) \sim n/\log n$. Dividing yields $p_n/\pi(n) \sim (\log n)^2$. Taking the floor changes the quotient by less than one, which is negligible relative to $(\log n)^2$. Thus $a(n)$ tends to infinity. If there were only finitely many strict increases, the sequence would be nonincreasing from some index onward and bounded above, a contradiction. Finally every descent requires $n+1$ prime. There are at most $\pi(N+1)-1$ such primes between 3 and $N+1$, inclusive. Apply the prime number theorem and divide by N. □

3 The remaining conjecture and attempted route

OEIS records the conjecture that there are infinitely many descents and credits the necessary prime-index condition to David A. Corneth (22 August 2020). The criterion above reduces the conjecture exactly to proving

$$r_n + (p_{n+1} - p_n) < \lfloor p_n/\pi(n) \rfloor$$

for infinitely many indices with $n+1$ prime, where r_n is the remainder on dividing p_n by $\pi(n)$. The prime number theorem controls the scale of the right side, but gives no adequate control of this remainder at those indices. Small gaps alone also do not bound the remainder. Thus neither the asymptotic nor a small-gap assertion completes this approach. No proof of infinitely many descents is claimed.

4 Scope and verification

These are supporting results and a reformulation of an unresolved assertion in this run. They are not presented as a solution of the OEIS conjecture or as novel theorems. Targeted searches of OEIS, arXiv, MathOverflow and Math StackExchange did not locate a solution during this run; this does not establish absence of one. ChatGPT generated and checked the deductions and tested the exact identity on a finite range; there has been no independent human review or formal verification.

References

[1] OEIS Foundation, *A111114*, definition, conjecture and D. A. Corneth's comment, `https://oeis.org/A111114`, consulted 11 November 2026.

[2] NIST Digital Library of Mathematical Functions, Sections 27.2 and 27.12, prime number theorem, `https://dlmf.nist.gov/27.2` and `https://dlmf.nist.gov/27.12`, consulted 11 November 2026.